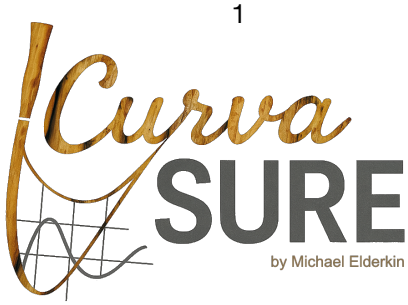


Real-time Double Reed Shaper 3D Modeling

CurvaSURE Shaper Generator [v5]

User Manual

Welcome to CurvaSURE! This program allows you to design and generate precise 3D printable straight shaper forms for various double reed instruments as well as exporting files that can be sent away to make metal versions. Additionally, CurvaSURE offers an enormous repository of well-established professional shapes that you can try, modify, or simply start from scratch to create your own!

1. Getting Started

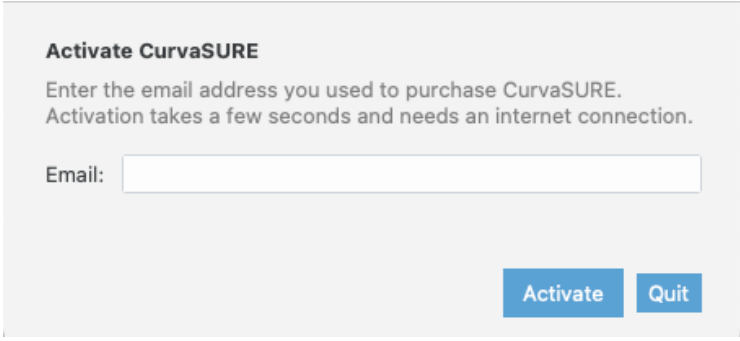
Upon launching the application, you'll see a main window divided into two sections:

- **Left Panel (Control Panel):** Contains all the input fields, parameters, and action buttons. This is where you define your shaper measurements.
- **Right Panel (Preview Area):** Displays a real-time 2D visual representation of your shaper as you adjust parameters.
- ⚙ **(Gear)**: Settings found in the upper left corner of the **Control Panel**. The menu contains **User Settings…** (see section 2.7), **Help** (opens this manual), **Show Licence** (your licence details and the licence agreement, see section 1.1) and **Contact Support**.
- **v5.0.0 :** The version number found in the upper right corner of the **Control Panel.** This informs you as to which version of CurvaSURE you are running.
    - **Auto-update:** CurvaSURE now has an auto update system which will alert you to a new version, if available, the next time you open the program. It will also include release notes for what has changed.
- **Activation:** The first time CurvaSURE runs on a computer it asks for the email address you used for your purchase. This only needs to be done once per computer. See section 1.1.

1.1. Activation and Licences

CurvaSURE is licensed to the email address used for your purchase and may be active on a limited number of computers at one time. The first time the program runs on a computer it needs to connect to the internet once to activate; after that it works offline.

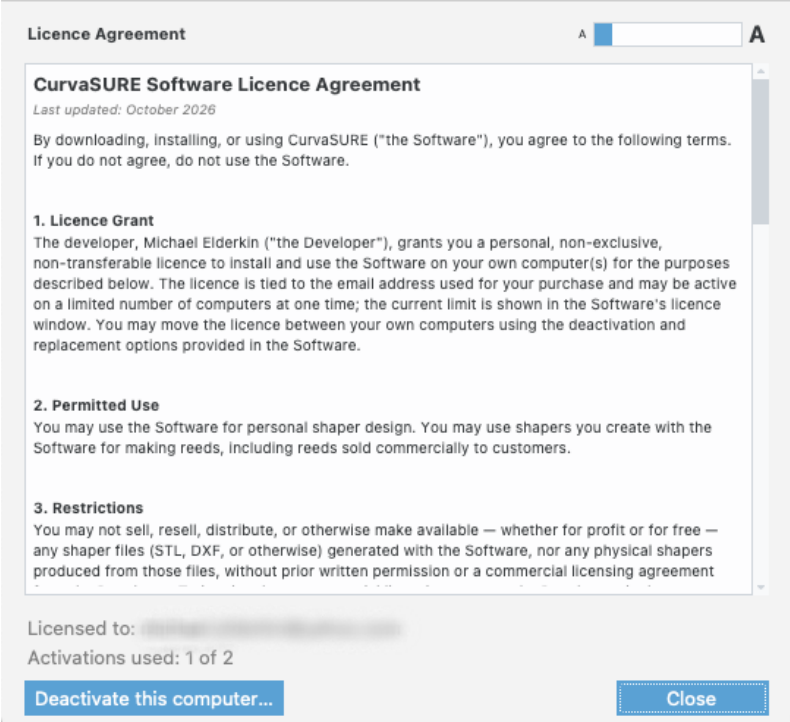
- **First launch:** When you open CurvaSURE you are shown the licence agreement first. After clicking **I Agree**, the activation window asks for your **purchase email**. Enter the email address you used when buying CurvaSURE and click **Activate**. If you close this window without activating, the program quits.



- **Activation limit:** Each licence may be active on a limited number of computers (the current limit is shown in the Show Licence window). If you activate on a computer after the limit has been reached, the activation window offers to **replace** one of your existing computers with this one. A replacement can only be made once in a while — if one was made recently, the message tells you in how many days another will be possible and suggests getting in touch.

- **Working offline:** Once activated, the licence is stored on the computer and CurvaSURE works without an internet connection.

- **Show Licence:** Click the ⚙ gear and choose **Show Licence** (on a Mac this is also in the **Help** menu). As well as the licence agreement, this window shows **Licensed to:** followed by your purchase email, and **Activations used:** followed by how many of your allowed computers are currently activated, for example “1 of 2”. If CurvaSURE has not been able to reach the licence server since it opened, this line reads “(connect to the internet to update)”.



- **Moving CurvaSURE to another computer:** In the Show Licence window click **Deactivate this computer…** and confirm. You will see “This computer has been deactivated. CurvaSURE will now close. Activate on your other computer with the same purchase email.” The program closes, the activation is freed, and you can activate the other computer by entering the same purchase email there. If a computer is no longer available to deactivate (lost, sold or reformatted), use the **replace** option described above when activating the new one.

- **Licence no longer active:** If a licence is deactivated remotely (for example after a refund or violation of terms), the program shows “CurvaSURE’s licence for this computer is no longer active. Please enter your purchase email to activate again, or quit.” If you believe this is an error, please get in touch (section 6).

- **Troubleshooting activation:**
    - While connecting, the window shows “Checking your purchase…”. If the licence server cannot be reached, CurvaSURE retries automatically (“Couldn’t reach the licence server — retrying in 5…”, then “retrying in 10…”), making up to three attempts over roughly 40 seconds.
    - If every attempt fails you will see “Could not reach the licence server. Check your internet connection and try again.” followed by “If you use a VPN or antivirus software, please try switching it off temporarily while the licence check happens, then switch it back on afterwards.” Some security software blocks the program’s first connection — switching it off just for the activation, then back on, usually resolves this.
    - “Could not verify the licence server’s certificate. This computer’s certificate store looks incomplete; please get in touch.” — this is a problem with the computer’s own security certificates rather than your connection. Please contact support.
    - A **Get in touch** link appears beneath any of these messages and opens the support page.
    - “Licence check failed” — “CurvaSURE could not check its licence on this computer. Please restart the program. If this keeps happening, contact support at …” Restart the program; if it keeps happening, get in touch.

Note: Activating sends your purchase email, an anonymised identifier for the computer, the computer’s name and the time of each check to the licence server — nothing else. No shapes or files leave your computer. See clause 7 of the licence agreement (⚙ → Show Licence).

1.2. Keyboard Shortcuts

Common actions have keyboard shortcuts. They act on the main window only and are ignored while a dialog is open.

Action	Mac	Windows
Undo	⌘Z	Ctrl+Z
Redo	⌘Y or ⌘⇧Z	Ctrl+Y or Ctrl+Shift+Z
Save Custom Preset	⌘S	Ctrl+S
Open Custom Preset	⌘O	Ctrl+O
User Settings…	⌘,	— (⚙ gear menu)
Quit (checks for unsaved work first)	⌘Q	— (close button)

2. Main Sections

2.1. Shaper Parameters

This section allows you to define the fundamental dimensions of your shaper.

- **Preset:**
    - **Select a Preset:** This dropdown menu provides a comprehensive list of pre-defined shaper profiles categorised by instrument (e.g., Bassoon, Historical Bassoon, Contrabassoon, Contraforte). Selecting a preset will automatically populate all relevant parameters and shape points, giving you a great starting point.
        - ***Online Presets:*** The application automatically checks for and downloads new or updated presets from an online repository each time you open the program. You can also manually trigger a check using the “Download/ Update Presets (Online)“ button in the Preset menu.

Preset names referencing well-known shaper profiles (e.g., Rieger 1a, etc.) are used solely to describe the style or geometry of the profile for user clarity. All measurements are either derived from publicly available information, independently taken, or sourced directly from the manufacturer.

These references are for identification purposes only and do not imply any affiliation with, endorsement by, or licence from the original manufacturers.

- **Open Custom Preset (⌘O / Ctrl+O):** Use this to load a previously saved **.json** preset file. If the shape you are working on has unsaved changes you will be asked whether to save it first (see section 2.8).
- **Save Custom Preset (⌘S / Ctrl+S):** Once you’ve designed a custom shaper you’re happy with, use this feature to save all its parameters, shape points and Notes to a **.json file**. This allows you to easily recall your custom design later or share it with others. The suggested file name is the shaper name as you typed it (only bracketed text and characters not allowed in file names are removed). After saving, the preset title changes to the name of the saved file — for example “Custom: Rieger 1A” saved as *Rieger 1A Alt.json* becomes “Rieger 1A Alt” — and the file then behaves like any other loaded preset. ***Important:*** *No custom presets are stored in the program memory, only locally on your hard drive.*
- **Restore Preset Backup…:** Every time you successfully generate files (section 2.6), CurvaSURE saves the exact shape — points, settings and Notes — to a hidden backup folder on your computer, so a shape you forgot to save can be recovered. Choose this item to see a list of backups with the **Date Generated**, **Shaper Name** and **File Name**, newest first. Select one and click **Restore** (double-clicking or pressing Return also works) to load it as the current shape, or **Close**. If you have unsaved work you will be asked about it first. Successful restores show “Restored ‘<name>’ generated on DD Mon YYYY at HH:MM.”; if

nothing has been generated yet the list reads “No backups yet — one is saved each time you generate files.” How long backups are kept is set in User Settings → Backups (section 2.7); identical repeat generations are not duplicated.

- **Download/Update Presets (Online):** Selecting this checks the online repository for any un-downloaded presets or updates to the presets that may have been published. The program will automatically check every time it opens, this is simply a way to manually engage that check.
- **Load Backup Database (Offline):** In the event that the program is unable to access the online repository, an offline backup can be loaded through this interface. Please get in contact if this is needed.
- **Clear Preset Cache:** This removes all the plugins cached on your computer. If you are having an issue with a preset or if you know a preset has been updated but doesn’t seem to be showing, click ‘Clear Preset Cache’ and then ‘Download/Update Presets (Online)’. This will download the newest versions of the presets into the program. (*This will not affect the presets you have exported and saved elsewhere.)*

- **Preset Search (🔍):** Clicking this opens the **Search Presets** window, which allows you to search the entire repository of shapes very easily by name or by the text of a shape’s Notes, and to filter by the measurements of the flat tip or the clipped flat tip. The window floats beside the main window, so you can keep working while it is open; clicking the 🔍 button again brings it to the front. Select a shape and click **Load Preset** (or double-click it) to load it.
  - ***Flat Tip:*** If you wanted a bassoon shape in the range of a 15.5 - 15.6mm flat tip, enter the minimum and maximum measurements with those values and all the shapes that fit within that range will appear.
  - ***Clipped Tip:*** If you normally clip your reed at 3mm and want a tip measurement of 15.5 to 15.7, you can enter all three values and see what appears. This can be useful to find similar shapes to experiment with.
  - **Notes column and panel:** A • symbol in the **Notes** column marks shapes that have Notes (click the heading to sort by it). Selecting a shape shows its Notes in the panel on the right (“Select a shape to see its notes.” until one is selected, or “No notes for this shape.”). See the Notes entry below.
  - **Keep Search Presets open after loading shape:** Tick this box at the bottom of the window to keep browsing after loading a shape. The setting is remembered between sessions. Click **Close** (or press Escape) to close the window.
- **Total Cane Length (mm):** This defines the overall length of the cane for which the shaper is designed. There are standards (such as bassoon being 120mm, contra being 150mm, etc) but ***any value can be entered in the text field between 50 and 180mm.***
- **Gouge Diameter (mm):** Specifies the diameter of the gouge to embed in the shaper. To be as precise as possible and to prevent your cane from being unnecessarily bent, it is generally recommended to select the correct gouge you use.

- ***Gouge Compensation:*** The program will automatically calculate the shaper width measurements needed to give you the correct shape based on the flat width. This means that if you change the gouge diameter, the shaper width numbers will change due to the cane wrapping around the shaper at slightly different dimensions.

- ***Note:*** *for gouges of* **16mm**, **18mm**, and **20mm** *(traditionally oboe and cor anglais), the concave shaper half uses a slightly smaller gouge (***14mm**, **16mm** *and* **17mm** *respectively) so that it grips the cane better. The convex half is still the set value.*

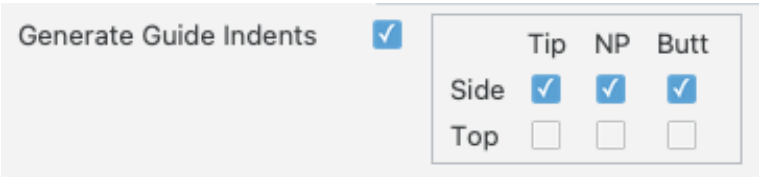
- **Bolt Hole Diameter (mm):** Sets the diameter for the bolt holes on the shaper which clamp the two halves together. The default value is **M5 Hex (5mm)** bolts, but there are more options for both metric and imperial measurements. There is an inbuilt **0.2mm** addition to this diameter to make threaded bolts slide through easier and to compensate for printing layers. This would make a 4mm hole actually 4.2mm. If ordering through Reeds 'n Stuff (section 2.6), a note reminds you that the M5 Hex option must be selected for shapers printed by them.

  - **Hex Bolts:** In general, hex bolts are recommended as there will be an indent created in the bottom of the shaper to nest it. This helps with tightening and doesn't require additional adhesion.

  - **Spring Indent:** As a default, there is a slightly larger sunken area (8mm in diameter) around each hole on the inside of both shaper halves. This is for spring placement, if you would like to make the shaper halves easier to use (see section 5.1 for the recommended springs). If springs are not used, these indents will not affect functionality.

- **Generate Guide Indents:** These controls place small physical indents at the tip **(Tip)**, narrowest point **(NP)** and butt **(Butt)** of the shaper, which serve as visual and tactile guides for accurate cane placement and shaping. The first tick box turns all indents on or off. Beside it is a small **placement matrix** with a column for each indent (**Tip / NP / Butt**) and a row for each position (**Side / Top**):

  - **Side** places the indent along the edge of the cutting surface, as a notch in the side of the shaper.



  - **Top** places it as a groove on both the top and bottom faces of the shaper.

  - Tick either box, or both, for each indent. Your preferred defaults can be set in User Settings (section 2.7).

  - In the preview, the NP and Butt side notches are drawn as a lighter blue overlay on the outline and Top grooves are drawn as thin rectangles across the shape.

- **Generate Alignment Posts (Recommended):** This checkbox controls whether alignment posts are embedded in the shaper to assist with cane placement. It is recommended this stay ticked as it will assist in both shaper alignment and strength.

- **Engrave Shaper Name (Recommended):** This checkbox controls whether the name of the preset is engraved on the outside face of the top shaper half (the base half carries “Created with CurvaSURE”). It is recommended to keep this enabled so as to easily identify the shaper after printing.

- **Shaper Name (Editable):** This box allows for customisation of the engraved shaper name, assuming *Engrave Shaper Name* is ticked. The box will auto-populate with the name of the loaded preset but it can be changed to whatever is desired. It is recommended to keep the character count, including spaces, to around 12 to maintain readability.

- **Notes:** Click the ▸ arrow beside **Notes** to open a free-text box where you can write anything about the shape — the maker, where the measurements came from, what it suits, wire positions and so on. Notes are part of the shape: they are saved inside the preset file, shown again whenever that preset is loaded (the box opens automatically if the shape has notes and stays closed if it doesn’t), and can be searched in Search Presets. Editing the notes marks the shape as *Custom:* like any other change, so the preset must therefore be saved again to retain any added notes.

2.2. Shape Points

This is where the true flexibility and power of CurvaSURE shines, allowing you to define highly customised shaper profiles. The shaper's profile is determined by a series of points (Distance, Shaper Width and Flat Width).

- **Folded Tip Width:** This shows the actual width of the reed tip when folded. When you form a blank, the gouge collapses into a flat line so the actual tip width of the blank will always be wider than a shaper that has the gouge built into it. This is helpful to better visualise the resulting tip opening.

- **"0 (Tip)":** This is the starting point of your shaper profile, representing the width at the very tip of the cane.

- **Intermediate Points:** These are user-defined points along the length of the shaper.

    - **Distance (mm):** This value specifies how far along the shaper's length (from the tip) the point is located.

        - **Accepted Range:** Values must be between **0mm and 95mm**.

    - **Shaper Width (mm):** This value defines the width of the cane at the specified distance from the tip of the reed.

        - **Accepted Range:** Values must be greater than **0mm and up to 30mm**, and at least **1mm narrower than the Gouge Diameter** (for 16, 18 and 20mm gouges, narrower than the smaller concave gouge — see section 2.1). Values too wide for the gouge turn red immediately.

    - **Flat Width (mm):** This value defines the width of the cane if the shape were flattened. If you have measurements from a fold-over shaper, these can be entered directly in this field and the corresponding straight shaper values will automatically be calculated.

- **Add Point:** Click this button to add a new intermediate point row. You'll enter a "Distance from Tip (mm)" and a "Width (mm)". To insert a row at a specific point, click in the field above where you want to insert a point, then click 'Add Point'.
- **Remove Point:** To remove an intermediate point, first **click on either the "Distance from Tip" or "Width" field** of the specific intermediate point you wish to remove. Once selected (the field will have focus), click the "Remove Point" button. You cannot remove the "Tip" or "Butt" required points.
- **Undo/Redo:** These buttons trigger the Undo/Redo function and are greyed out when there is nothing to undo or redo.

- **"— (Butt)":** This is the final point of your shaper profile, representing the width at the butt end of the cane. If you have a blank template selection, its "Distance from Tip" is automatically calculated as half of the "Total Length (mm)". If you have selected a preset and not a template, the '(Butt)' text will not appear.

Input Validation: The application performs real-time validation of your input values. If a value is outside the accepted range or in an invalid format, the input field will turn red, and an error message will appear in the preview area or via a pop-up.

2.3. Curve Options

This section controls how the application interpolates between your defined shape points.

- **Smooth Curve (Recommended):** This option uses a cubic spline interpolation method to create a smooth, continuous curve between your defined points. This is generally recommended for most shaper designs as it produces a more organic and flowing profile. To get the benefit of the curve, multiple points are needed. Generally speaking, the more points you have the better your shape will turn out.
  - As you're inputting points, you may notice that the Smooth Curve option creates some bizarre shapes. This is only the mathematics trying to make sense of the overall shape of the points currently available. As you work your way down the shaper, the wildness of the curves will be tamed - you simply need more data points.
  - ***Tip:*** When creating your own shape from scratch, choose the **Straight Lines** option. This will help you sketch the general shape out. Once you have enough points, switch to the **Smooth Curve** option and edit from there.
- **Straight Lines:** This option connects your defined points with straight line segments. This can be useful for creating more angular or segmented shaper profiles, particularly with fewer Shape Points. This is also useful for experimenting with 'rough' or 'experimental' shapes prior to engaging the smooth curve, as you can input fewer points and still have a workable result.

If you have enough points listed, the differences in the Smooth Curve and Straight Line settings become very small to negligible.

2.4. Advanced Features

- **Offset Shape Start:** This slider allows the start of the shape to be pushed back or pushed up a specified amount from the tip, while retaining the proportions of the original shape. The slider covers **±5mm**; to offset further, type a value of up to **±15mm** into the box beside it — the slider then sits at its end. Touching the slider while it is pinned at a value past 5mm snaps the value back to the slider position.
    - ***Example:*** If you wanted to retain the tip width of the shape after clipping 2mm off the tip, you would offset the shape start by 2mm.
    - ***Example:*** If you are using a shape that was a foldover originally, you might want to set the offset to the value your cane sticks up above the tip of the foldover shaper. This is usually around 0.5-0.7mm. That way you will have a more accurate shaping result to the original shape.
    - This setting *will* be saved if you export a preset, including values typed beyond the slider's range.
- **Global Width Change:** This slider allows you to change the overall width of the shape by the amount set on the slider. You might find yourself wanting a slightly wider or narrower shape overall - this dialogue allows for you to experiment without having to go through and manually adjust the shaper values.
    - When you engage the slider with a positive or negative value, options will appear below to apply the width change to either the Shaper Width or the Flat Width. **The change will apply to whichever option is selected** - you do not need to click the Apply button below unless you want to permanently change the points.
    - The button **Apply Width Change to Shape Points** should only be clicked if you want the width change to be reflected in the actual shape points.
    - ***Note:*** When exporting a shape as an STL, DXF or saving a preset - **the width change will be maintained** even if you've ***not*** pressed the apply button. This allows for saving a shape and experimenting while keeping the integrity of the original measurements.
- **Enable Asymmetrical Shape:** This tick box allows the user to adjust the left and right side of the shape independently. Once you tick the box, you will see that the Left Profile and Right Profile tabs become active and two more buttons appear in the Shape Points area: **Load Preset as Side** and **Mirror to Opposite Side**. Enabling asymmetry on a loaded preset marks it as *Custom:*.
    - **Load Preset as Side:** When you toggle this, you may go up to the preset menu and select a preset (internal or loaded custom). The values of this preset will then be loaded into whichever profile tab you have selected (Left or Right). This function will remain active until you select a preset or you click the button again, which should currently display **Cancel Load as Side**.
    - **Mirror to Opposite Side:** When you click this button, the tab that is currently displayed will copy the values it contains to the opposite side. For example, if you have the Left Profile selected, the values would then be copied into the Right Profile.

- When you **disable** asymmetry, the program uses the Left Profile as the points to construct the symmetrical shape.

- **Display Half Width Values:** This tick box becomes available to use when Enable Asymmetrical Shape is ticked. This changes the values in the Shape Points dialogue to half their widths. This is helpful in creating asymmetric shapes as the half widths will be the distance from the midpoint line to the outside edge.

- **Generate Shaper Ears:** Ticking this box will make ears appear around the tip. On many oboe and cor anglais fold-over shapers, this is a popular feature, providing extra cane around the tip when shaping. The default width is 0.5mm and length is 7mm, but these values can be changed (**maximum 0.5mm width** and **5-10mm length range**).

- **Scale Shape:** Clicking the ▸ arrow beside **Scale Shape** opens the scaling dialogue. This feature allows the user to select any shape and scale it up or down in any way they might like.

  - **How to Scale:** Select your desired shape (for example, a bassoon shape) and open **Scale Shape**. As bassoon shapes are usually 120mm, you should see 120mm in the **mm total** box. If you then want to scale this shape to contrabassoon size (usually 150mm), you would then enter 150mm in that box and hit apply. You will notice that the tip width (the **mm tip (flat)** box) in the scale dialogue changes value when you change the length - reflecting the scaled up size. If you decide that the scaling caused the shape to be too narrow or wide, you can untick the **Lock ratio** box and the Width and Length scaling are able to be used independently.

    You can use any of the four boxes to scale the shape (if the ratio is locked) but it might be helpful to start with the Length (mm total) box, as you can be certain it will scale to your desired length. All other boxes will adjust to whichever values are input. (It's much easier to know that a bassoon shape scaled to contrabassoon should be 150mm in length than it is to know the ratio is 1.25!)

  - ***Note:*** Scaling will not occur unless you click **Apply Scale to Shape**.

  - ***Note:*** The scaling ratio will always apply to the base shape that is loaded. If you scale a bassoon shape to a contrabassoon shape (1.25x) and you want to change it back, simply enter 1.0 in the ratio or change back to the original length. Hitting **Apply Scale to Shape** more than once won't cause compounding scaling.

  - **Scaling Exclusion:** Enter two distances (mm from the tip, measured on the original shape) in the **Exclude from … to … mm** boxes to leave that part of the shape unscaled — for example to keep a bocal/crook or reamer fit. **Transition** blends the scaled shape smoothly into the excluded region over that many mm on each side. With **Lock Excluded Region Position** ticked, the excluded region stays at exactly the same mm-from-tip position as on the original shape; unticked, the region keeps its size but its start moves to the appropriate position based on the scale of the other points (e.g. 10mm → 12mm at ×1.20).

- **Populate Points:** Clicking the ▸ arrow opens the Populate Points dialogue. There are two fields where you enter a range along the shaper and a third field for the frequency of measurement. Clicking the Populate Points button samples the curve as it is currently drawn and adds a shape point at each interval across that range, without changing any

existing points. This is a quick way to turn a shape defined by a few points into one with points at a specified interval before fine-tuning.

- ***Note:*** If working on an asymmetric shape, you must Populate Points independently for each side.

2.5. Preview Controls

The right panel provides a visual preview of your shaper.

- **Display Preview using Flat Width:** Ticking this box will display the preview as if the shape was completely flattened. This is purely a visual change, helping the user to visualise what a completely flattened piece of cane would look like. *This does not affect the file generation.*
- **Show Guide Marks:** Toggles the display of helpful visual guides on the preview:
  - **Red Dashed Line:** Represents the centre line of the shaper.
  - **Green Dashed Line:** Mark the butt end of the shaper.
  - **Orange Dashed Line:** Indicate the narrowest point of the shape.
  - **Purple Dashed Line:** If Offset Shape Start is engaged, this indicates the place to where the tip measurement is offset.
- **Show Point Marks:** Ticking this will display small black dots along the shaper outline. This is primarily to help determine how the points are affecting the overall shape, particularly if the Smooth Curve option is enabled.
- **Show Midpoint Line:** Ticking this will display a dotted line down the centre of the shaper. This is particularly helpful for visualising asymmetrical shapes.
- **Auto-update:** When enabled, the preview will automatically regenerate in real-time as you modify parameters or shape points.
- **Generate Preview:** If "Auto-update" is disabled, click this button to manually refresh the preview.
- **Show Clipped Tip at __ mm:** Ticking this box and entering a value will display a dotted line at the specified distance from the tip. This line represents where you will clip the reed after forming. The distance may be anywhere on the blade, from **0** (the tip itself) up to **the butt end** (half the Total Cane Length), with an absolute limit of **90mm**. Values off the shape turn the field red and show “Clipped Tip distance must be on the shape: between 0 and NN.N mm for this blade length.” (or “Clipped Tip distance cannot be more than 90 mm.”); if no cane length has been set yet, “Set the Total Cane Length before placing the clipped tip.” If the shape is later shortened or scaled so that the mark falls off it, the field turns red and the mark (and the blade area below) are hidden until a valid value is entered.

- An annotation box will appear labeled **Clipped Tip**. Inside this box will be the **distance** from the tip, the **shaper width** at the clipped tip, and the corresponding **flat width**.

- **Show Collar at __ mm:** Ticking this box and entering a value will display a collar line at a specified distance from the tip. This can help visualise how your profile will look on the shaper. The same limits apply as for the clipped tip (0 to the butt end, maximum 90mm), with matching “Collar distance …” messages.
  - As above, an annotation box will appear with distance, shaper width and flat width values.
  - Ticking both **Clipped Tip** and **Collar** will **calculate the total blade area** between the collar and the tip and appear in the **Clipped Tip** annotation box. If the clipped tip is behind the collar, or both sit on the same spot, the box shows “Area: (Invalid Range)” instead.

- **Show Curvature Comb:** The Curvature Comb is a diagnostic tool that helps you visualise the smoothness of your shaper's profile when using the "Smooth Curve" option. When activated, it displays a series of lines, or "teeth," perpendicular to the shaper's outline. The length of each tooth is directly proportional to how sharply the shaper is curving at that point. Longer teeth indicate a tighter curve, while shorter teeth show a flatter, gentler curve. A smooth and gradual transition in the length of the teeth signifies a well-formed curve without any unwanted bumps or kinks. A perfectly straight line should have no teeth visible.
  - You can adjust the comb using the controls in the "Preview Controls" section. The **Comb Density** slider changes the number of teeth shown, providing a more or less detailed view of the curve. The **Comb Scale** slider acts as a magnifying glass, increasing or decreasing the length of the teeth. Increasing the scale can help exaggerate and identify very subtle imperfections in the curve that might otherwise be hard to see, allowing you to fine-tune your points for a perfectly smooth shaper outline.

- **Load Preset as Reference:** Ticking this box will enable the Load Preset as Reference functionality, which allows for two shapes to be overlaid and compared with one another.
  - Tick the box to enable, click **Load Preset as Reference** and go to the menu, selecting any preset as you normally would. The preset will then be loaded as a reference shape behind the current shape. To switch between shapes, click between **A** and **B** either in the Preview Controls or in the upper right of the display area, beneath the pan controls.
  - The **A shape** is always the shape **you are working on** and the **B shape** is the **reference**. Whichever shape is selected will be shown as the normal **blue outline** in the front with the other shape as a dotted **orange outline** behind it.
  - Ticking **Show Clipped Tip** and **Show Collar** will display different annotation boxes for each shape, allowing you to compare the widths and area for each shape.
  - Hovering your mouse over the shape will **display the widths for both the working and the reference shape for easy comparison.**

- To disable the background shape, either untick the box (to hide the shape) or click **Clear Reference**, which removes that preset from the reference field.
- Custom presets opened from a file can also be loaded as a reference, including asymmetrical ones. If a file contains no usable shape points you will see "Reference Not Loaded — '<name>' does not contain shape points that can be used as a reference."

- **Additional Preview Controls**
  - **Zoom / Pan *(upper right corner of the preview area)*:** These are buttons which allow you to increase the size of the preview as well as moving around. The **Pan** controls are arrows that allow you to move around the image. The central circular button will both center and zoom out the image. There are **Zoom** and **Pan** buttons which allow you to increase the size of the preview as well as moving around. The **Pan** controls are arrows that allow you to move around the image. The central circular button will both center and zoom out the image.

    Note: You can also zoom using your mouse scroll wheel and pan by grabbing the shaper/grid and moving it around.
  - **Distance, Width, and Width from Midpoint Measurements:** Additionally, if you **hover your cursor over the outline** you will be able to see the distance and width at any point along the shaper. If you have an **asymmetric preset loaded**, the **Width from Midpoint** value will be displayed as well.

2.6. Action Buttons

- **Generate Selected Files:** This is the core function for exporting your shaper design.
  - Clicking this button first builds the files — a **progress bar** replaces the button while this happens and the window is busy until it finishes — and then prompts you to choose a save location:
    - With **both STL and DXF** ticked, one dialog titled "Save Shaper Files (a folder with this name will be created)" appears. The two files are saved together inside a new folder of the name you choose, as *Name/Name.stl* and *Name/Name.dxf*. If a file of that name already exists in that location you will see "Cannot Save — A file called 'Name' already exists there, so a folder of that name cannot be created. Please choose another name."; if the folder already contains shaper files, "Replace Existing Files?" lists them and asks before overwriting.
    - With **one format** ticked, the dialog is titled "Save Shaper File" and saves a single file. The name is kept exactly as typed — only a trailing .stl or .dxf is treated as an extension, so *shaper.v2* becomes *shaper.v2.stl*.
    - The suggested file name is the Shaper Name. On success you will see "Files saved successfully" followed by the saved paths, and a backup of the exact shape is written for **Restore Preset Backup…** (section 2.1).
  - The application can generate two types of files. Tick either or both to generate:

        - **STL (.stl):** A 3D model file suitable for 3D printing. The generated STL contains the two shaper halves side by side, ready for printing. Each half is 17mm tall as standard (a particularly deep gouge cut makes them slightly taller — you are told when this happens). The base half carries the alignment posts, the hex-nut recess and the "Created with CurvaSURE" engraving; the top half carries the alignment holes (**4.4mm**, a close fit on the 4mm posts) and the shaper name. Which half is the base — the concave half (**Smile**) or the convex half (**Frown**) — is set in User Settings → Gouge Orientation (section 2.7). The two 8mm spring/bolt counterbores are the same depth on both halves (3.8mm each, 7.6mm in total), and by default a **0.8mm chamfer** removes the sharp ridge where the gouge meets the side wall of the concave half (User Settings → Chamfer). Guide indents are cut into the side profile and/or engraved as grooves on the top and bottom faces according to their placement (section 2.1).

        - **DXF (.dxf):** A 2D CAD file representing the outline of your shaper, useful for laser cutting or further CAD work.

    - **Error Handling:** The application validates all parameters before generation. If any required parameters are missing, values are invalid or red, the shape is too wide or narrow for the gouge, or the Total Cane Length is out of range, an error message explains what to fix and nothing is generated. If the 3D geometry itself cannot be built — for example a gouge channel, the convex dome, the bolt or alignment holes, the hex pocket or the engraved text fails to cut cleanly — you will see an **Export Error** naming the step that failed and suggesting a slightly different width, gouge or cane length. CurvaSURE no longer substitutes simplified geometry silently in these cases, so a file that saves is a file that is correct.

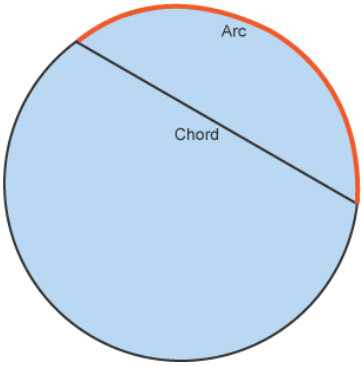
    - **Saved with warnings:** Non-fatal notes are reported after a successful save in a "Saved with warnings" dialog listing the files followed by "Please note:" and the points — for example "The two halves will be slightly taller than the normal shaper height due to a deeper gouge cut being needed.", or that the concave edge chamfer could not be applied on one side (leaving that edge sharp).

- **Help:** Links back to this very manual!

- **Shaper Measurement Converter *(Legacy Tool)*:**

  [As of CurvaSURE v3.0.0, this is no longer necessary due to the added Flat Width column.]

  This is a computational tool that will convert measurements between straight shapers and fold-over shapers or flat reed measurements. Straight shapers maintain the gouge radius while you shape the cane whereas fold-over shapers force the cane at the tip flat (or almost flat). This tool accounts for the differences in curvature of the cane, so if you wanted to input the measurements of a fold-over shaper, you would get the correct straight shaper values.



Example: Let's say you wanted to convert a Rieger 1A shaper tip to a straight shaper. On the Rieger website (and confirmed by measuring the actual shaper tip) the tip width is **15.5mm**. If you shape a piece of cane using this tip, the resultant tip will be 15.5mm as the fold-over shaper more or less resembles the folded state of the reed.

However, if you were to apply those measurements to a straight shaper with no adjustment, then the resulting measurement of the folded tip would be an incorrect width of **16.18mm** (using a 32mm gouge).

This is because the **chord length** (the width measurement of a piece of cane measured from above or the straight shaper width) was used instead of **arc length** (the total length of the flattened out tip).

We want a consistent tip measurement regardless of our gouge settings, therefore the **arc length** *must* stay the same and the gouge will be the same. This only leaves the **chord length** variable, so to achieve an accurate shape transfer it is this value that needs to be adjusted.



Therefore, if you have a shaper tip you would like to convert into a straight shaper, take measurement values at the tip and every 5-10mm, plug those values into the fields provided along with the desired straight shaper gouge, and press **Calculate**. Add these values to the Shape Points field and voilà, you have a straight shaper copy!

- If there is a slight curvature at the back of the shaper tip, you can choose to take that as a flat-measurement and accept the slight resulting width difference (often less than 0.1mm) or you can take a radius gauge measurement at various increments and plug different values in. For example, if the back of a shaper tip has a very slight bend (i.e. 40mm radius), then plug in 40mm as the gouge and take the overhead measurement and enter that value into the Straight Shaper field. Convert and get the Flat Measurement. Next, erase the Straight Shaper value and change the Gouge value to match your desired shaper (while keeping the Flat Measurement you had) then generate the new Straight Shaper value. It's a roundabout process to do it this way, but you're guaranteed to convert your shaper correctly!

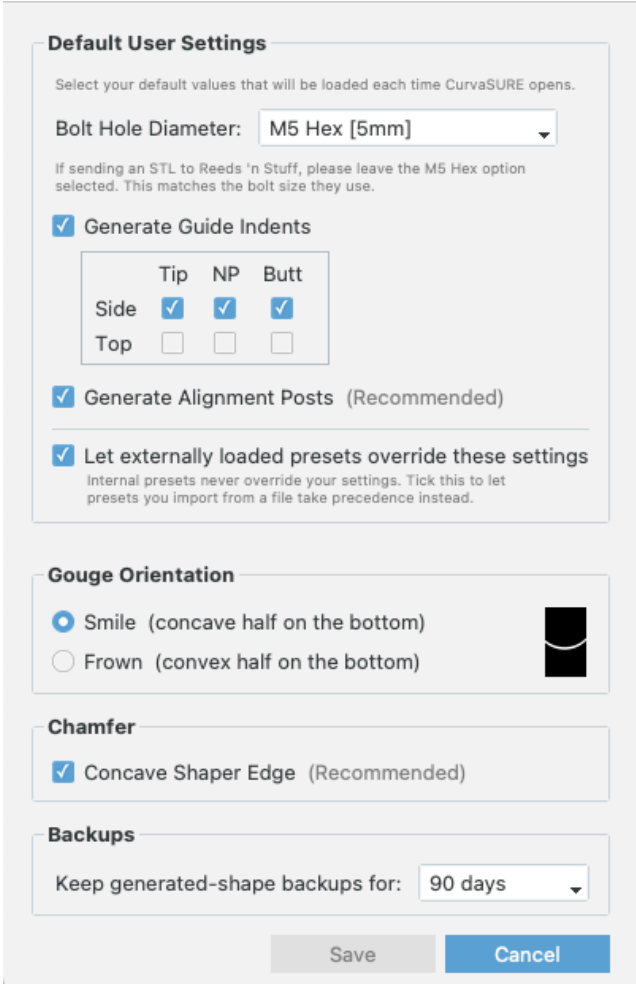
- **Order at Reeds n' Stuff:** Clicking this button will direct you to the Reeds n' Stuff website, displaying the CurvaSURE order page. On this page you will be able to order 3D printed shapers, if you don't currently own a printer (exporting the **STL** file). You will also be able to order metal versions of the shapers you designed (exporting the **DXF** file).

    - ***Note:*** Before ordering, please check what gouge settings, length, and bolt holes you are using. In order for Reeds n' Stuff to accurately print or machine your shapers, you must use the settings they mention.

2.7. User Settings

Open **User Settings…** from the ⚙ gear menu (on a Mac also from the application menu, or press ⌘,). These are your own "house style" for the computer you are using: the settings in the top group are loaded every time CurvaSURE opens and are re-applied over whatever a preset specifies, so an internal preset can never change your tooling choices. The settings below that affect how the shaper is printed and are never stored in presets. **Save** is only enabled once something differs from the saved settings; **Cancel** (or Escape) closes without saving.

- **Default User Settings** — “Select your default values that will be loaded each time CurvaSURE opens.”
    - **Bolt Hole Diameter:** the bolt hole that every loaded shape will use (see section 2.1).
    - **Generate Guide Indents** with the same **Tip / NP / Butt × Side / Top** placement matrix as on the main panel.
    - **Generate Alignment Posts** *(Recommended)*.
    - **Let externally loaded presets override these settings:** “Internal presets never override your settings. Tick this to let presets you import from a file take precedence instead.” This is on by default, so a custom preset you open from a file (or restore from a backup) brings its own bolt hole, indents and posts with it; untick it if you always want your own defaults, whatever the file says.



- **Gouge Orientation: Smile (concave half on the bottom)** or **Frown (convex half on the bottom)**. This changes which half of the printed shaper is the base: the Smile and Frown names describe what the shaper looks like when sitting upright on a flat surface, and an illustration in the dialog shows each. The base half carries the alignment pins, the hex-nut recess and the “Created with CurvaSURE” text; the other half carries the alignment holes and the shaper name. Choosing **Frown** therefore moves the pins, hex recess and “Created with” text to the convex half and the holes and shaper name to the concave half. Choose this before generating files (see section 5.3). This is a setting for this installation of CurvaSURE and is not exported with presets.

- **Chamfer — Concave Shaper Edge** *(Recommended)*: on by default. Adds a 0.8mm chamfer along the top edge of the concave shaper half, where the gouge meets the side wall, taking the sharp, fragile ridge off the edge. For quality of printing it is recommended to keep this ticked; otherwise you may have to do light sanding on the concave side once printed. This setting is *not* exported with presets.

- **Backups — Keep generated-shape backups for:** 30 days, 60 days, 90 days or Keep all (default 90 days). Every time you generate files the preset for that exact shape is saved to a hidden folder so it can be recovered from **Restore Preset Backup…** (section 2.1). Older backups are deleted after this many days, when CurvaSURE starts and when these settings are saved; at most 1,000 backups are kept whatever the setting.

2.8. Unsaved Work

CurvaSURE keeps track of whether the shape on screen has been saved, and asks before anything would replace it. The prompt appears when you **close the program** (including ⌘Q on a Mac), and when you **switch to another preset** — from the Preset menu, Search Presets, Open Custom Preset or Restore Preset Backup. It offers **Save**, **Don't Save** and **Cancel**; choosing Save opens the usual Save Custom Preset dialog, and cancelling that save keeps you on the current shape.

- For a modified preset the message reads "You have unsaved changes to '<name>'. Do you want to save your preset before closing?" (or "…before loading '<next>'?" / "… before switching presets?").
- Shapes built from a blank or **(Default)** template are protected too. As soon as you have entered a width on either side or written any Notes, the prompt reads "The shape you are building from '<template>' has not been saved. Do you want to save it as a preset before …?" Choosing a cane length or gouge on its own does not count as work. Once the shape has been saved, the template no longer prompts until it is changed again.
- Loading a preset **as a Reference** or **as a Side** never prompts, because neither replaces the shape you are working on.

3. Tips for Best Results

- **Validate Inputs:** Always ensure your numerical inputs are within the accepted ranges. Invalid inputs can lead to unexpected or unprintable shapes.
- **Start with 'Straight Lines' Enabled:** Due to the nature of how the spline curve works, the first values entered can make a strange visual. If you enable **Straight Lines** first and then sketch out your general shape, you'll have a lot of success when engaging the **Smooth Curve** option.
- **Preview Regularly:** Even with auto-update, occasionally review the preview to ensure the shape is developing as intended.
- **Understand Your Cane:** The ideal shaper profile depends heavily on the specific cane you are using (its thickness, density, and natural curve). Experimentation and iterative design are key.
- **3D Printing Considerations:** When generating STL files, remember that the final print quality will depend on your 3D printer's capabilities and slicer settings. Ensure you use appropriate materials and print resolutions.

4. Troubleshooting

- **"Input Error" / "Plotting Error":** Check the values you've entered in the "Shaper Parameters" and "Shape Points" sections. Look for red-highlighted fields.

- **"Cannot Remove Point":** Remember you can only remove intermediate points, not the "Tip" or "Butt" required points. Ensure you've clicked on an intermediate point's field to select it before attempting to remove.

- **“Invalid Shape Data” / “Invalid Cane Length”:** Generating files or saving a preset is blocked while any shape point is red or the Total Cane Length is outside 50–180mm. The message lists the offending points; correct the highlighted fields and try again.

- **“Shape Too Wide” / “Shape Too Narrow”:** The shape runs over the width limit for the selected gouge (1mm less than the gouge, or than the smaller concave gouge for 16/18/20mm), or below the 3mm minimum width, somewhere along its length — including between points, where a smooth curve can overshoot. The preview shows the affected stretch in red and “Shape Points are too wide for the Gouge, please change either parameter.” (or “…too narrow, please change or add more points.”). Reduce the width, add a point to tame the curve, or choose a larger gouge.

- **“Export Error”:** The 3D geometry could not be built: the message names the step that failed (the gouge channel, the convex dome, the bolt or alignment holes, the hex pocket, the engraved text, or a layer that is not a closed solid). This is rare and usually depends on an unlucky combination of width, gouge and cane length — change one of them slightly and generate again. If it keeps happening with a particular shape, please send it to support (section 6). Review all fields, especially those related to “Total Cane Length”, “Gouge Diameter” and “Bolt Hole” dimensions.

- **“Cannot Save”:** When saving STL and DXF together, a folder with the chosen name is created. If a *file* of that name already exists in that location the folder cannot be made — choose another name. “Replace Existing Files?” simply asks before overwriting files already inside that folder.

- **Activation problems:** “Could not reach the licence server…”, “Could not verify the licence server’s certificate…”, “Licence no longer active” and “Licence check failed” are all covered under *Troubleshooting activation* in section 1.1. The most common cause of a failed activation is a VPN or antivirus program blocking the first connection.

- **“Reference Not Loaded”:** The custom preset file you tried to load as a reference has no usable shape points (for example a template with empty widths).

- **No Logo/Icon:** If the logo or icon doesn't appear, it simply means the image files are not found in the expected location. The application will still function normally.

- **"Connection Error" / "Data Error" for Presets:** This indicates an issue connecting to the online preset server or a problem with the downloaded data. Check your internet connection.

- **“Duplicate distance values are not allowed”:** This means that in your shape points, there are two width values listed at the same distance. The program doesn’t know which value takes priority, so one will need to be removed. ***Note:*** if you have changed the total cane length of a template shape, you might unintentionally get a duplicate value if the new butt distance (which changes automatically to match the length) overlaps with a previously existing value. The same rules apply here - one value will need to be changed, but the selected value will need to go in the (Butt) field.

5. Materials, 3D Printing Tips, Assembly and Shaping

5.1. What You Will Need

Below you will find the materials needed for fully assembling the straight shaper.

- **3D Printer:** Any sort of 3D printer will do, as long as the printing bed is large enough to be able to print the Total Length parameter. The Bambu Lab A1 or A1 Mini is a great choice to start off with, if it happens to be in your price range.

- **Filament:** In testing 3D printed shapers out, the best performing filaments of the group were **PLA**, **PLA+**, and **PETG**.

    - **PLA/PLA+:** PLA is a very commonly used filament praised for the high tensile strength it provides and evenness of printing surface. PLA+ is a modified version of this meant to be slightly stronger and more flexible. Printing with PLA/PLA+ is a great choice to start off with, but be aware that it can flake if you dig a knife into it. Care needs to be taken when shaping to prevent digging into the filament surface. Sandpaper is a useful tool to smooth areas out you can't reach with a knife.

    - **PETG:** This is a filament with much higher flexibility than PLA/PLA+ but it is much more resistant to knife scraping. Due to the strength it possesses, it is often used for manufacturing parts that go through wear and tear. Depending on the infill percentage you end up using, you may find that if you can twist it ever so slightly but this is not an issue when the other half is attached and tightened. ***Warning:*** this filament is naturally much 'stickier' than PLA/PLA+ so you may have to clean the hot end of your printer more often or possibly purchase a hot end with a different tip.

  Any of these are good options, so it is worth experimenting to find out which you prefer.

- **Bolt and nut (at least 40mm in length)**: For clamping the two halves together, a bolt and nut are needed. The appropriate length for this is a minimum of 40mm as it will reach though the shaper giving just enough threading on the other side for the nut to close around. The bolt size needs to match the Bolt Hole Diameter option that you chose when generating the shaper.

    - ***Recommended:*** Print a Hex Nut Grip which can house the nut meant to screw the bolt in. This is much easier to tighten and handle than using the nut on its own. Please click **here** to download the recommended STL files. After printing two of them, insert the nut into the bottom of the print and screw into the bolt. *Note:* if you have calipers, then it's a good idea to measure the flats of the bolt against the inside of the Hex Nut Grip to see if it needs to be scaled. A tighter fit is better!

- **Compression Spring 0.6 x 7 (or 8) x 15mm (optional)**: For additional ease of keeping the shaper open, a compression spring may be used. The shaper has embedded indents around the bolt hole where these springs may be placed. The indent on each half is 8mm in diameter, therefore any spring with that diameter will fit. However, the recommended height of the spring is 15mm. This will allow the spring to compress fully when closed without too much effort and have enough force to push the two halves

apart when the bolts are loosened. Any spring thicker than 0.6mm will also be difficult to push closed.

- **Steel Shim Tape [30 microns or similar] (optional):** Using steel shim tape prevents your knife from cutting into the shaper filament at all, effectively transforming it into a metal shaper - at least in functionality. See section 5.4 for more information.

- **Digital Calipers (optional):** To easily measure reeds and also test the output of your prints, having digital calipers is highly recommended.

5.2. 3D Printing Tips

As straight shapers are fairly complex geometrical shapes, the question may arise how to get the best result printing them. Luckily there are only two options and both of them have their own merits. Experimentation to find out which works best for you is the way to go, but pros and cons of both options will be listed below based on experience.

- **Print Settings**: The following print settings are recommended for the best result.
    - **15% *infill, Gyroid or Grid*** - Gyroid is suggested as it won't cross over itself while printing.
    - ***Wall thickness*** - *3* layers at 0.4mm (1.2mm thickness in total)
    - ***Avoid crossing walls*** - If your printer has this function, then it is highly recommended that you engage it so as to prevent and filament build-up and inconsistencies in the print if the nozzle catches on a wall. This is especially important if you print vertically, as the shapers can easily topple.
    - ***Z seam*** - If your slicer has this option, select **Back** as the placement for the Z seam. This moves the position of where the layer printing starts to the end of the shaper. Many printers will start in the middle of the shaper around the tip, which can lead to inaccurate results.
    - ***Layer Height*** - As fine as your printer can do. Smaller layers will generally yield better accuracy.

5.3. Assembly

Once you have all your materials and printed halves, it is time to assemble the shaper.

- **Step 1**: The orientation of the shaper — whether it 'smiles' or 'frowns', referencing the way the gouge is pointing when the shaper sits on a flat surface — is chosen in **User Settings → Gouge Orientation** *before* the files are generated (section 2.7). The half you chose as the base has the alignment posts, the hexagonal nut recess and the "Created with CurvaSURE" text; the other half has the alignment holes and the engraved shaper name and goes on top. The default, and recommended, orientation is **Smile** (concave side on the bottom and convex side on the top) as it is much easier to rest the cane in when positioning. With this orientation, the engraved shaper name faces up.

- **Step 2**: Take your bolts and insert them through the bolt holes from the bottom up through the top. The head of the bolt should be on the bottom of the shaper and the threaded ends should be sticking up through the top.
    - If you selected Hex Bolts in the program, there will be a hexagonal indent printed on the base half of the shaper which allows for the bolt to fit within the shaper and not need additional fastening when opening or closing.
    - If you do *not* have a Hex Bolt and have a rounded top, then dab a bit of super glue around the top of the bolt, underneath the head, and then press and hold into the bottom shaper half until fastened. This is useful if you don't want the bolt to spin around while you are tightening the shaper.
    - If you choose not to glue it, when you tighten the nut on top place your thumb on the bolt head to prevent it from turning.
- **Step 3 (optional)**: If using springs, take the top half of the shaper off and drop a spring around each one of the threaded bolts sticking upwards. The springs should fit neatly into the indents on the inside of the shaper. Place the top half back on, being careful to line up the springs with the indents on that half.
- **Step 4**: Tighten a nut around each one of the upwards facing bolt threads, making sure that the shapers can close all the way.

5.4. Shaping

Due to the nature of 3D printing filament, there may be some concern over what using an abrasive, such as a knife or sandpaper, might do to the final shape. Based on testing, there hasn't been change to the overall dimensions of the shaper even after repeated uses.

This comes down to a few things:

- Using PLA+/PETG filament which seems to be more abrasive-resistant
- A shallow angle when cutting along the shaper (5-10 degrees above the shape)



If you take care while shaping to keep that small angle, then you will find it won't catch on the shaper.

Fail-safe solution: There is a method that will allow you to shape without worrying about cutting into the shaper at all. This method utilises **steel shim tape**. If you are able to source around 30 micron (0.03mm thick) shim tape (8mm wide), it can be placed on either side of the shaper cutting area to protect the filament underneath.

This only adds a total of 0.06mm to the shaper width which is a functionally negligible addition, so the shape should not need to be altered.

Tip: If you do want to be precise, you can remove the total combined thickness of the tape from the shape using the Global Width Change. In the example above, you would move the slider to -0.06mm and make sure it is being applied to the **shaper width**, as this is where the thickness from the tape is being offset.

A recommended brand is **Matrix Engineering *Super Shim Plus (0.03mm).*** This is a British brand and may not be available near you, so any comparable thickness / brand should work. If you are US-based, a similar brand is **Cutting Edge Grinding & Supply Co *Super Shim (.001”)***.

6. Support

Any support inquiries or suggestions for features can be address directly via this form.

If the form isn’t working or if you haven’t heard back, please reach out via this website.

The activation window and the licence error messages also carry a **Get in touch** link which opens the same support page.

Thank you and enjoy CurvaSURE!